

Differentiated Carbon Prices or International Transfers?

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However **efficiency requires a uniform price**, and equity concerns can be settled by **transfers**.

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We compute the emissions rights equivalent to prominent differentiated prices proposals.

Differentiated prices *vs.* a uniform price

Why differentiate? Four imperfections

Absent imperfections, a uniform price with transfers is optimal (Aldy & Stavins, 2012). LICs allocated more rights than emissions: they clearly gain.

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⇒ Inconsistent with survey evidence (Fabre et al., 2025; Fabre, 2026).

None of the four decisively discard a uniform price: 1. uncalibrated, 2. irrelevant, 3. compatible, 4. unproven.

Standard Theory

The static case

Country i produces y_i , consumes c_i , and at carbon price p_i faces abatement cost a_i :

$$c_i = y_i - a_i + t_i, \quad t_i = \text{international transfer.}$$

Welfare u_i depends on c_i and on **global emissions** $E = \sum_j e_j n_j$, **fixed at E^***

n_i is population, e_i emissions p.c.

p^* : uniform price attaining E^* , i.e. **p^* s.t. $E^* = \sum_j e_j(p^*) n_j$.**

Shorthands $e_i^* := e_i(p^*)$, $a_i^* := a_i(e_i^*)$.

A **situation** for i is described by $s = (t_i, p_i)$. Two anchors:

Benchmark $s^* = (0, p^*)$: uniform price, no transfer.

Autarky $s^a = (0, p_i)$: differentiated price, no transfer.

u_i can be expressed as function of s : we have $u_i(0, p_i) = c_i^a = y_i - a_i$, and $u_i(t, p^*) = c_i^* + t = y_i - a_i^* + t$.

Step 1: the equivalent transfer

$V_i(p_i)$: the lump sum transfer which makes i as well off with the global price as in autarky:

$$u_i(V_i(p_i), p^*) = u_i(0, p_i) \quad \implies \quad V_i(p_i) = c_i^a - c_i^* = a_i^* - a_i.$$

Optimising country sets $\frac{da_i}{de}(e_i^*) = -p^*$: Therefore, a first-order Taylor expansion around e_i^* gives:

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Equivalent transfer g_i^s of an arbitrary situation $s = (t_i, p_i)$, relative to the benchmark:

$$g_i^s = \underbrace{t_i}_{\text{cash received}} + \underbrace{V_i(p_i)}_{\text{gain from facing } p_i \neq p^*}$$

Step 2: the correspondence between prices and rights

Cap-and-trade at p^* , each person in i endowed with rights r_i . Net monetary transfer to i :

$$t_i^* = (r_i - e_i^*) p^* \quad \implies \quad g_i^* = t_i^*.$$

Autarky, from Step 1: $g_i^a = V_i(p_i) \approx (p^* - p_i) e_i^* \varepsilon^*$. Setting $g_i^* = g_i^a$:

$$\begin{aligned} & \boxed{(r_i - e_i^*) p^* \approx (p^* - p_i) e_i^* \varepsilon^*} \\ \iff & \quad r_i \approx \left(1 + \left(1 - \frac{p_i}{p^*}\right) \varepsilon^*\right) e_i^* \quad \iff \quad p_i \approx \left(1 + \frac{e_i^* - r_i}{e_i^* \varepsilon^*}\right) p^*. \end{aligned}$$

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The key formula for policy design: rights r_i under uniform price equivalent to autarky price p_i

Low autarky price ($p_i < p^*$) $\Rightarrow$ more rights than emissions $r_i > e_i^*$ to compensate the extra abatement cost.

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$\Rightarrow$ Any differentiated-price arrangement maps onto a cap-and-trade allocation.

If the two can always be made equivalent, *is there any reason to prefer one?* Yes.

Step 3: the uniform price weakly dominates, for *every* country

Give country i rights equal to its *autarky* emissions, $r_i = e_i$. Net transfer $\tilde{t}_i = (e_i - e_i^*)p^*$.

Convexity of abatement costs implies:

$$a_i - a_i^* \geq (e_i - e_i^*) \frac{da_i^*}{de} = -(e_i - e_i^*)p^* \quad \implies \quad \tilde{t}_i \geq a_i^* - a_i = V_i(p_i) = g_i^a.$$

Hence, consumption increases *in every country* when trading is allowed and prices equalize:

$$\tilde{c}_i = y_i + \tilde{t}_i - a_i^* \geq y_i - a_i = c_i^a$$

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This is [Diamond & Mirrlees \(1971\)](#) production efficiency applied to climate: absent other distortions, **direct transfers beat differentiated factor prices as a redistributive instrument.**

The dynamic case: the equivalence breaks

Add time t ; each variable becomes a trajectory. β_t = discount factor, $R_i = \sum_t r_{it} = i$'s carbon budget.

$G_i = \sum_t g_{it} \beta_t$: total discounted equivalent gain

Welfare equivalence now requires equality of *entire discounted streams*, $G_i^U = G_i^A$:

$$\sum_t \beta_t (r_{it} - e_{it}^*) p_t^* \approx \sum_t \beta_t (p_t^* - p_{it}) e_{it}^* \varepsilon_t^*.$$

In the static case p^* was a *common factor* and cancelled. Here it does not: the timing of allocation matters.

A **generous** allocation in a **low-price** period is **worth less** than the same allocation when prices are high.

⇒ **No direct correspondence between the intertemporal carbon budget R_i and welfare.**

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Special case 1 — homothetic prices, fixed emission rights shares: $p_{it} = \pi_i p_t^*$ and $r_{it} = \rho_i r_t$

$$\mathcal{B} := \sum_t r_t \beta_t p_t^* \quad \rho_i \mathcal{B} - \sum_t e_{it}^* p_t^* \beta_t \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^* p_t^* \beta_t,$$

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Analogous to static formula, but $\mathcal{B}$ still requires the full price path ⇒ **not tractable without a model.**

Special case 2: Hotelling — and why it is the right benchmark

Hotelling rule: Fixed aggregate carbon budget + perfect foresight + efficient capital markets
 $\Rightarrow$ carbon price rises at the rate of return on capital. Discounting at that same rate, $\beta_t p_t^* = p_0$ is constant.

$\beta_t p_t^*$ then factors out of both sums, and the dynamic condition collapses to:

$$\boxed{R_i - E_i^* \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^*} \quad E_i^* = \sum_t e_{it}^*$$

Structurally identical to the static formula: the intertemporal carbon budget R_i plays the role of the single-period allocation, and the homothetic price factor π_i the role of p_i/p^* .

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In the numerical analysis, we place ourselves in the Hotelling case.

Numerical analysis

Taking the formula to a calibrated model

The formula rests on approximations a calibrated model need not satisfy: homogeneous elasticities, homothetic price paths, a first-order Taylor expansion.

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Grid search over (ρ_i, π_i) for eight major regions: **USA, Russia, EU27, China, Turkey, India, Nigeria, DRC**.

The two scenarios

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Revenue recycled domestically under Negishi weighting (transfers $\propto$ inverse marginal utility)
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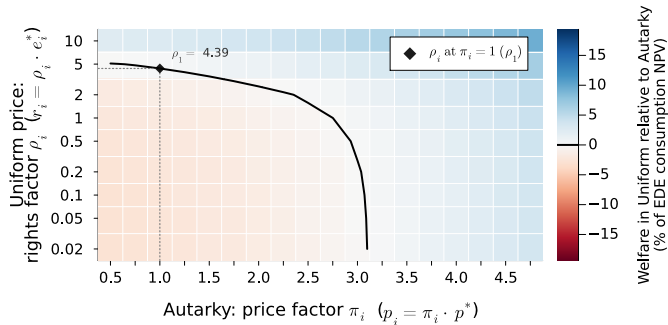
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RoW receives the *residual* cap, grandfathered on baseline emissions.

Reading the heatmaps

USA welfare **gain of uniform over autarky**, depending on rights and price.

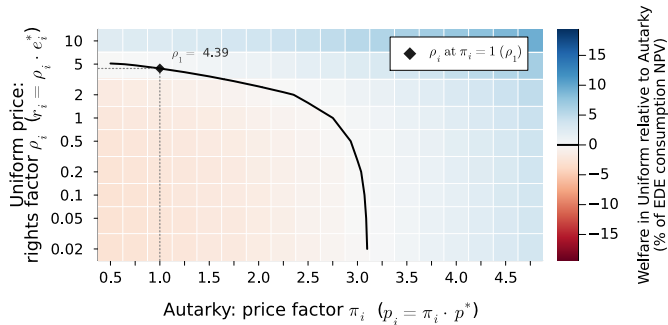


Grid showing the welfare **gain** or **loss** of *Uniform over Autarky*

Blue: Uniform price with transfers dominates; **Red:** Autarky dominates; line: indifference curve.

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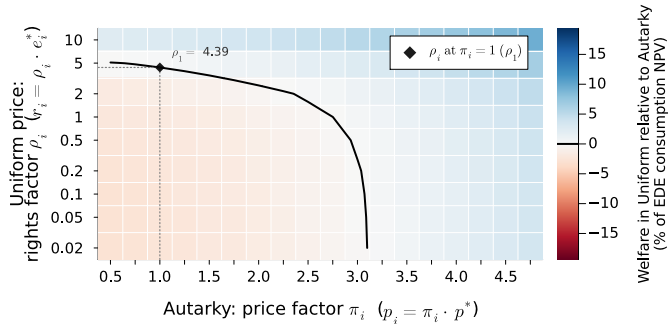
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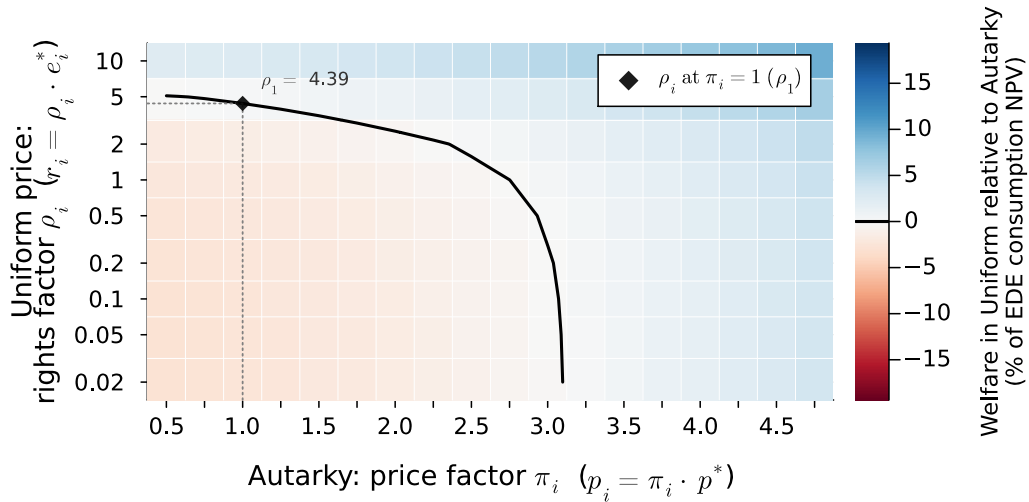
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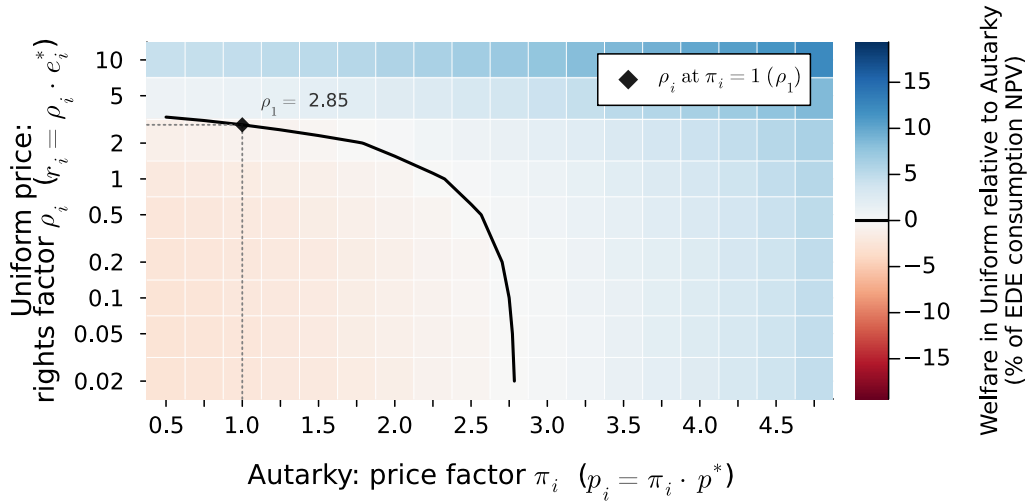
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Reading: The U.S. would need at least $\rho_1 = 4.39$ times more rights p.c. than the world average to be as well as off as in autarky, with the same global price.

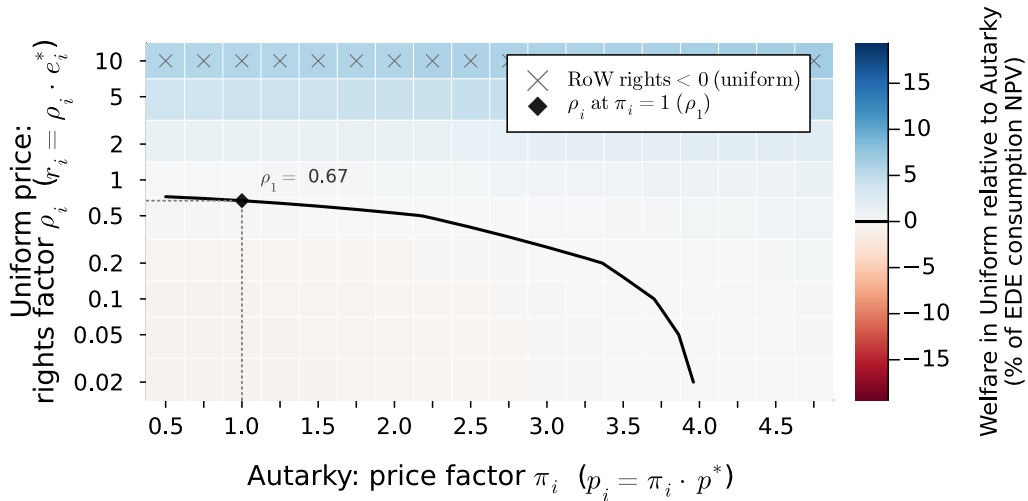
United States



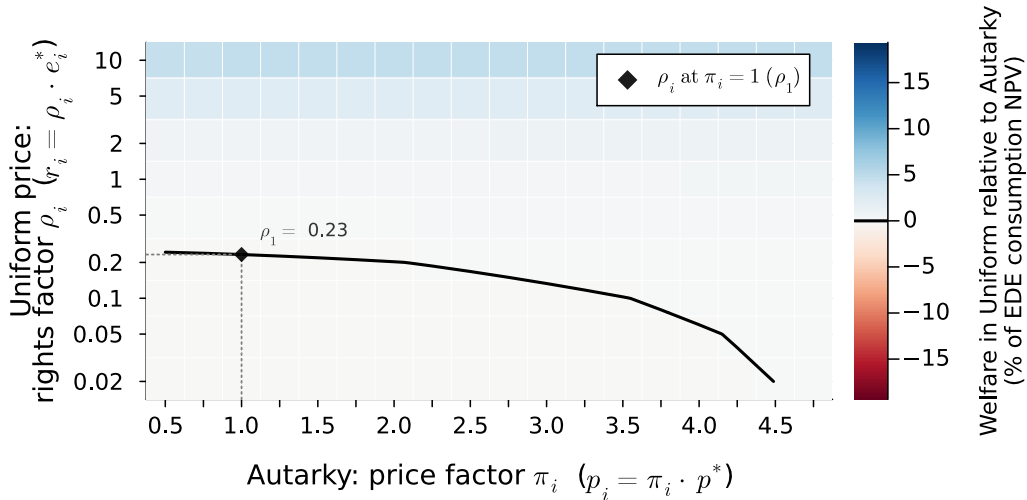
Russia



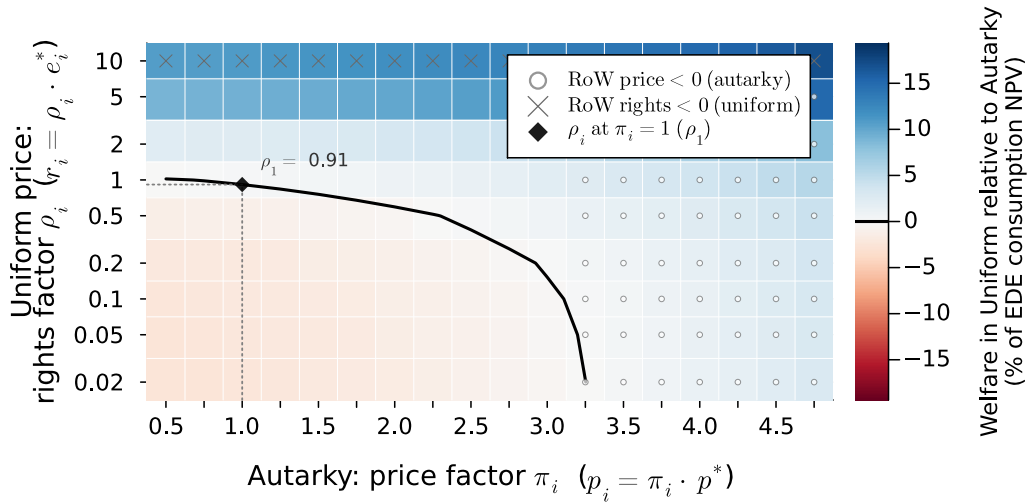
China



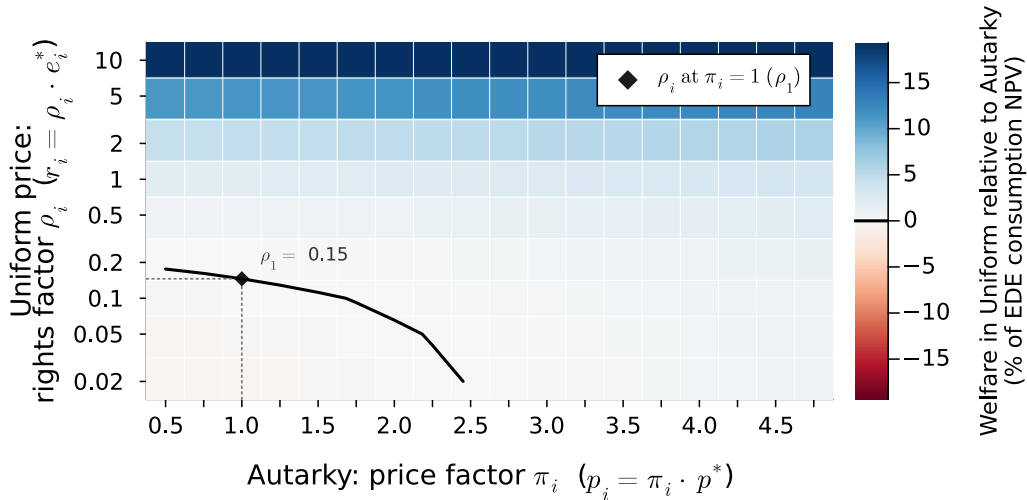
European Union (EU27)



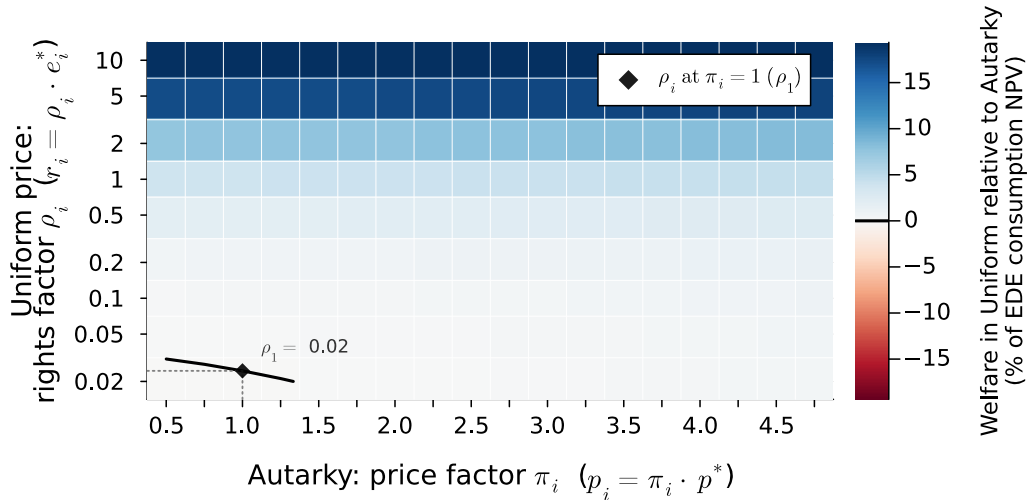
India



Nigeria



Democratic Republic of the Congo



Assessment of live proposals

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We price HICs at \$75/t (Wolfram's top tier): unpriced HICs would make the “bargain” a cut in world

Equivalent rights for the proposals

Country	Wolfram et al.			Banerjee et al.		
	p_i	ρ^A	ρ^B	p_i	ρ^A	ρ^B
United States	0	—	—	75	2.16	2.82
Russia	0	—	—	75	1.36	1.89
European Union	75	0.26	0.36	75	0.26	0.36
Turkey	0	—	—	50	1.54	1.62
Indonesia	50	0.89	0.92	50	1.09	1.04
China	50	0.84	0.98	50	0.88	0.95
Brazil	50	0.38	0.37	50	0.5	0.41
India	25	1.3	1.12	30	1.18	1.03
Nigeria	0	—	—	30	0.24	0.11
DR Congo	0	—	—	10	0.06	0.01
Gain in rights, variant 1 (%)		9.9%	9.5%		9.1%	8.5%
World temp. 2100, change (°C)		−0.020	−0.019		−0.037	−0.034
World welfare gain, EDE (variant 2)		0.028%	−0.006%		0.102%	−0.009%
World consumption gain (variant 2)		0.022%	0.018%		0.057%	0.048%
World welfare gain, EDE (variant 3)		0.028%	0.011%		0.102%	0.041%
World consumption gain (variant 3)		0.022%	0.020%		0.057%	0.047%
World welfare gain, EDE (variant 4)		0.081%	0.048%		0.419%	0.298%
World consumption gain (variant 4)		0.024%	0.021%		0.068%	0.059%

The comparison regime prices the proposal's own members at a single price (\$49.2/t Wolfram et al., \$55.8/t Banerjee et al. in 2030) and leaves non-members unpriced, as the proposal does. p_i : the price the proposal asks of i in 2030 (\$/t); a country it does not price is shown at $p_i = 0$ with no equivalent allocation ('—'). ρ_i : allocation as a multiple of an equal-per-capita share of the club's emissions; ρ^A solves each country alone, ρ^B all jointly (distribution solved to 0.03–0.05%, level held at option A's). Variant 1 lets the rights set the cap, so the temperature row is the change from the proposal's own 2100 warming (2.16°C Wolfram et al., 1.80°C Banerjee et

Conclusion

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A **first-best statement**. The dominance result assumes away every other distortion. Where the second-best arguments bite — market power, pre-existing sectoral taxes, divergent discount rates (Anthoff et al., 2022; Babiker et al., 2004) — differentiation can be optimal in its own right, and we are **silent**. ▶ The three arguments

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The dynamic equivalence leans on Hotelling. Imperfect foresight or frictional capital markets break the constancy of $\beta_t p_t^*$ and **reopen the timing problem** the static formula cannot see. ▶ Why Hotelling

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Thank you for your attention!

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Appendix

Two boundary conditions evaluated in *every* year of 2030–2100:

1. **Market-distorting subsidies** ($\omega_i \times \pi_i > 1$): the target country claims more than 100% of the global price burden. To compensate, the model is forced to push $p_{-i} < 0$ — economically, a **fossil-fuel subsidy for the RoW**.

2. **Negative emission rights** ($\rho_i > 1 / pop_i$): granting i its multiplier uses more than the whole global budget, forcing the RoW into **negative rights**. Binds for large populations: countries above 10% of world population (China, India) trigger it for any $\rho_i > 10$.

RoW price identity, the starting point of the root-finding:

$$p^* = \omega_i p_i + (1 - \omega_i) p_{-i}, \quad \omega_i = e_i / E^* \quad \implies \quad p_{-i} = p^* (E^* - e_i \pi_i) / e_{-i}.$$

This linear approximation assumes emission-weighted price averaging delivers the same global abatement — false with non-linear MACs, hence the **Newton–Raphson** correction to $< 0.1\%$ relative error on the cap.

Autarky scenario, United States ($\pi = 2$) [Go back](#)

Variable	2030	2040	2050	2060	2070	2080	2090	2100
p^* (USD/tCO ₂)	42.6	112.9	205.9	285.7	401.6	480.9	476.2	471.4
p_i (USD/tCO ₂)	85.2	225.8	411.7	571.4	803.3	961.9	952.3	942.8
E_i (GtCO ₂)	4.31	2.58	0.63	-0.53	-1.75	-2.22	-1.86	-1.52

Net-negative US emissions from the 2060s: abatement rates $\mu_i > 1$, i.e. [CDR deployment](#), with the MAC function extrapolated beyond the 100% boundary.

Standard IAMs constrain $\mu_i \in [0, 1]$; enforcing that bound crashes the grid search under extreme policy parameters.

The equivalent price, and why we do not use it [▶ Go back](#)

One could instead define an **equivalent price** p_i^* such that $(0, p_i^*)$ is welfare-equivalent to (t_i, p_i) :

$$t_i + (p^* - p_i)e_i^*\varepsilon^* \approx (p^* - p_i^*)e_i^*\varepsilon^* \quad \implies \quad p_i^* \approx p_i - \frac{t_i}{e_i^*\varepsilon^*}.$$

Problem: p_i^* can be *negative* when the transfer is large, which has no natural interpretation as a carbon price.

The equivalent transfer g_i^s does not share this limitation $\implies$ we use it as the primary welfare metric.

Last slide (needed)